

Arjuna JEE 2025

Maths

DPP: 1

Quadratic Equations

- Q1** Find the value of a for which $(a - 2)x^2 + bx + c = 0$ is a quadratic equation.
 (A) for all $a \in R$
 (B) for all $a \in R - \{2\}$
 (C) for all value of a
 (D) for $a \in \{2\}$
- Q2** The value of x satisfying $\log_3(x^2 + 4x + 12) = 2$ are
 (A) 2, -4
 (B) 1, -3
 (C) -1, 3
 (D) -1, -3
- Q3** If one root of the equation $3x^2 - 10x + 3 = 0$ is $\frac{1}{3}$, the other root is
 (A) $-\frac{1}{3}$
 (B) -3
 (C) 3
 (D) $\frac{1}{3}$
- Q4** The roots of $x^2 - (1 + \sqrt{2})x + \sqrt{2} = 0$ are
 (A) Real and equal
 (B) Real and unequal
 (C) Imaginary
 (D) None of these
- Q5** The value of p for which $x^2 + 5px + 16 = 0$ has no real root, are:
 (A) $p > \frac{8}{5}$
 (B) $p < \frac{-8}{5}$
 (C) $\frac{-8}{5} < p < \frac{8}{5}$
 (D) None of these
- Q6** A value of k for which the quadratic equation $x^2 - 2x(1 + 3k) + 7(2k + 3) = 0$ has equal roots is
 (A) 1 (B) 2
 (C) 3 (D) 4
- Q7** If the roots of $x^2 - bx + c = 0$ are two consecutive integers, then $b^2 - 4c$ is
 (A) 0 (B) 1
 (C) 2 (D) None of these
- Q8** If the product of the roots of the equation $(a + 1)x^2 + (2a + 3)x + (3a + 4) = 0$ be 2, then the sum of roots is
 (A) 1 (B) -1
 (C) 2 (D) -2
- Q9** If α and β are roots of quadratic equation $x^2 - 4x + 2 = 0$, then value of $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} =$
- Q10** The value of 'a' for which one root of the quadratic equation $(a^2 - 5a + 3)x^2 + (3a - 1)x + 2 = 0$ is twice as large as the other is
 (A) $-\frac{1}{3}$
 (B) $\frac{2}{3}$
 (C) $-\frac{2}{3}$
 (D) None of these
- Q11** Difference between the corresponding roots of $x^2 + ax + b = 0$ and $x^2 + bx + a = 0$ is same and $a \neq b$, then



- (A) $a + b + 4 = 0$ (B) $a + b - 4 = 0$
 (C) $a - b - 4 = 0$ (D) $a - b + 4 = 0$

Q12 If $(1 - p)$ is a root of quadratic equation $x^2 + px + (1 - p) = 0$ then its root are-

- (A) 0, -1 (B) -1, 1
 (C) 0, 1 (D) -1, 2

Q13 If one root of the equation $x^2 + px + q = 0$ is the square of the other, then

- (A) $p^3 + q^2 - q(3p + 1) = 0$
 (B) $p^3 + q^2 + q(1 + 3p) = 0$
 (C) $p^3 + q^2 + q(3p - 1) = 0$
 (D) $p^3 + q^2 + q(1 - 3p) = 0$

Q14 If $\tan \alpha$ and $\tan \beta$ are the roots of $x^2 - px + q = 0$, then find the value of $\sin^2(\alpha + \beta)$.

- (A) $\frac{q^2}{p^2 + (1 - q)^2}$
 (B) $\frac{q^2}{p^2 + (1 + q)^2}$
 (C) $\frac{p^2}{p^2 + (1 + q)^2}$
 (D) $\frac{p^2}{p^2 + (1 - q)^2}$

Q15 If the difference between the roots of the equation $x^2 + ax + 1 = 0$ is less than $\sqrt{5}$, then the set of possible values of a is

- (A) $(3, \infty)$
 (B) $(-\infty, -3)$
 (C) $(-3, 3)$
 (D) $(-3, \infty)$

Q16 The number of real roots of the equation $(x - 1)^2 + (x - 2)^2 + (x - 3)^2 = 0$ is

- (A) 1 (B) 2
 (C) 3 (D) None of these

Q17 The least integral value of k for which the equation $x^2 - 2(k + 2)x + 12 + k^2 = 0$ has two distinct real roots is

- (A) 0 (B) 2
 (C) 3 (D) 4

Q18 The value of k for which $2x^2 - kx + x + 8 = 0$ has equal and real roots are

- (A) -9 and -7
 (B) 9 and 7
 (C) -9 and 7
 (D) 9, -7

Q19 $x^2 + x + 1 + 2k(x^2 - x - 1) = 0$ is a perfect square for how many values of k

- (A) 2 (B) 0
 (C) 1 (D) 3

Q20 If α and β are the roots of the equation $x^2 - 4x + 1 = 0$ the value of $\alpha^3 + \beta^3$ is

- (A) 76 (B) 52
 (C) -52 (D) -76

Q21 If the roots of the equation $12x^2 - mx + 5 = 0$ are in the ratio 2 : 3, then $m =$

- (A) $5\sqrt{10}$
 (B) $3\sqrt{10}$
 (C) $2\sqrt{10}$
 (D) $10\sqrt{5}$

Q22 The roots of the given equation $(p - q)x^2 + (q - r)x + (r - p) = 0$ are

- (A) $\frac{p-q}{r-p}, 1$
 (B) $\frac{q-r}{p-q}, 1$
 (C) $\frac{r-p}{p-q}, 1$
 (D) $1, \frac{q-r}{p-q}$

Q23 The value of λ such that sum of the squares of the roots of the quadratic equation, $x^2 + (3 - \lambda)x + 2 = \lambda$ has the least value is -

- (A) 1
 (B) 2



- (C) $\frac{15}{8}$
(D) $\frac{4}{9}$



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Answer Key

Q1 (B)
Q2 (D)
Q3 (C)
Q4 (B)
Q5 (C)
Q6 (B)
Q7 (B)
Q8 (B)
Q9 6
Q10 (B)
Q11 (A)
Q12 (A)

Q13 (D)
Q14 (D)
Q15 (C)
Q16 (D)
Q17 (C)
Q18 (D)
Q19 (A)
Q20 (B)
Q21 (A)
Q22 (C)
Q23 (B)



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